

FreshRoute — Food Bank Distribution Optimizer (India)

Continuous Evaluation-1: Data Preparation & Analysis Report

Real India APMC arrivals + prices (DMI, 28 states, 2705 markets, 304 commodities) | 383,753 records | Surplus-glut triage label (High / Medium / Low) | Reproduce: `python scripts/clean_dataset.py`

September 2026 | Group members: (fill names/roll nos. before submission) | Time: 10-minute viva — every member presents

1. Project Title & Overview

FreshRoute predicts a **redistribution priority (Low / Medium / High)** for every APMC mandi-commodity-day so volunteers move the most urgent glut first. Built only on real, cited, India-native mandi records (no US proxy, no invented donation logs). The priority label is **derived by a published glut-proxy rule** and disclosed as such — it is a triage aid, not measured waste. Output of CE-1 is a zero-missing, ML-ready dataset with a frozen X/y contract plus 6 EDA figures.

2. Problem Statement

Edible surplus rots at mandis while nearby shelters face shortages because nobody ranks **what should move first**. Mandis see daily volume gluts that crash wholesale prices — an observable glut signal. Fixed first-come-first-served rules ignore the perishability × need trade-off. ML helps by learning a 3-class triage function from arrival-surge × price-drop patterns, so dispatchers get a ranked move-first list instead of raw arrivals.

3. Objectives

1. Curate a real, cited, India-native dataset (383k records).
2. Deliver zero-missing, encoded, ML-ready data with a frozen X/y contract.
3. Derive a disclosed triage label (Surplus Intensity S + saved quantile thresholds).
4. Produce 6 EDA figures with observations that justify features and expose bias.
5. Enable an India pilot design (mandi → priority → ranked hub).

4. Dataset Collection & Description

Source: Directorate of Marketing & Inspection (DMI), Ministry of Agriculture & Farmers Welfare — data.gov.in API 9ef84268-d588-465a-a308-a864a43d0070 (Current Daily Price of Various Commodities from Various Markets) / CEDA Ashoka Agmarknet mirror. License: Govt Open Data License–India (free academic reuse). File: `data/raw/apmc_arrivals_prices.csv` (81 MB, git-ignored; re-download steps in `data/SOURCES.md`).

File	Shape	Role
raw APMC	394,258 x 22	Untouched download: mandi ops, 2025-10-27 to 2025-11-19 (24 days)
cleaned	383,753 x 22	This pipeline; 0 missing; keeps market names + geo for ranking
encoded	383,753 x 62	ML-ready: 61 X cols + y

Key column	Type	Meaning
date	date	Trading day
state / district / market_center	text	Where the glut sits (28 states, 528 districts, 2705 markets)
commodity / variety / grade	text	What the glut is (304 crops; perishables dominate)
latitude / longitude	float	Market pin (37% imputed, flagged)

Key column	Type	Meaning
arrival_tonnes	float	Volume arriving that day (median 0.68 t, highly skewed)
min / max / modal_price	float	Rs./quintal → standardised to Rs./kg (/100)
price_z / arrival_z / surplus_S	float	Engineered: deviation from 6-day market baseline; S = arrival_z x (-price_z)
redistribution_priority	High/Med/Low	DERIVED triage label from S quantiles

Coverage note: 24-day window gives breadth (markets × crops) but no seasonal patterns — stated openly; WFP long-baseline prices are planned for CE-2.

5. Data Cleaning

Issue	Action (logged in cleaning_report.json)
Exact duplicates	Checked; 0 found (not assumed)
Text / dates	Strip + UPPERCASE state/market/commodity; date → datetime; 4 rows missing key IDs dropped
Units	Rs./Quintal → Rs./kg (/100). 10,503 Bundle/Unit rows dropped — not convertible; logged
Impossible values	Negatives in arrivals/prices → NaN → removed; logged
Geo gaps	140,784 rows (37%) missing lat/lon → state median + geo_missing flag
Variety / grade gaps	1 each → UNKNOWN
Outliers	Per-commodity IQR winsorize: price k=3, arrivals k=5 (keeps real shocks, trims typos)
Temporal order	Sort market → commodity → date before rolling baselines

6. Missing Values — Identify, Treat, Verify

Column	Missing (raw 394k)	Treatment
modal_price / date / market IDs	4 rows	Drop (essential identifiers)
min_price / max_price	141 / 161	Kept; modal used for features
latitude / longitude	147,858 (37%)	State median + geo_missing flag (spatial signal kept)
variety / grade	1 each	UNKNOWN
After pipeline	0 total	Asserted in script: cleaned.isna().sum() == 0

7. Categorical Data Encoding

One-Hot (drop_first=True, int) for low-cardinality fields only; high-cardinality market/district (2705/528) stay in cleaned for ranking but out of X to avoid explosion. Target label-encoded only: High→0, Low→1, Medium→2 (saved in report). No scaling in CE-1 (deferred to CE-2).

Field	Encoding	Columns
state_name (28)	One-Hot	27 dummies
commodity_top (top-20 + OTHER)	One-Hot	20 dummies
market / district / variety / grade	Excluded from X	Kept in cleaned for analysis/ranking
redistribution_priority (y)	Label map	High 0 / Low 1 / Medium 2

Result: X = 61 cols (14 numeric/flags + 47 dummies) | y = redistribution_priority. Contract: train_test_split(stratify=y, random_state=42) in CE-2.

8. Exploratory Data Analysis & Visualization

6 frozen figures regenerated by scripts/clean_dataset.py plus 3 supplementary bias/operations charts (Figs 7–9) into reports/figures/. Target split is usable but imbalanced — stratify in CE-2.

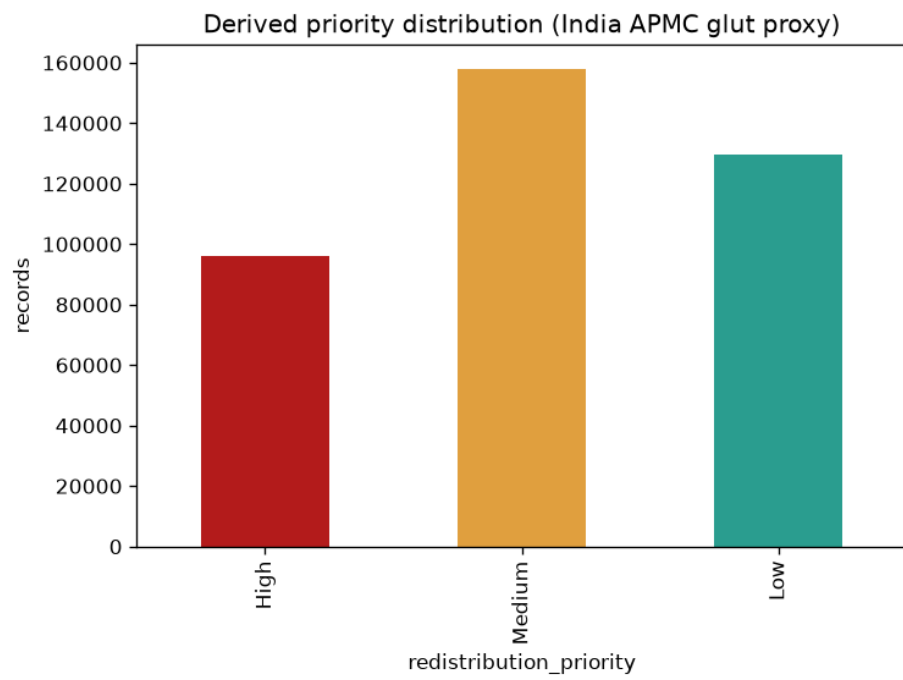


Fig 1 — Derived priority: Medium 158,026 / Low 129,788 / High 95,939. Observation: usable 3-class split, mild imbalance → stratified splits + F1-macro in CE-2.

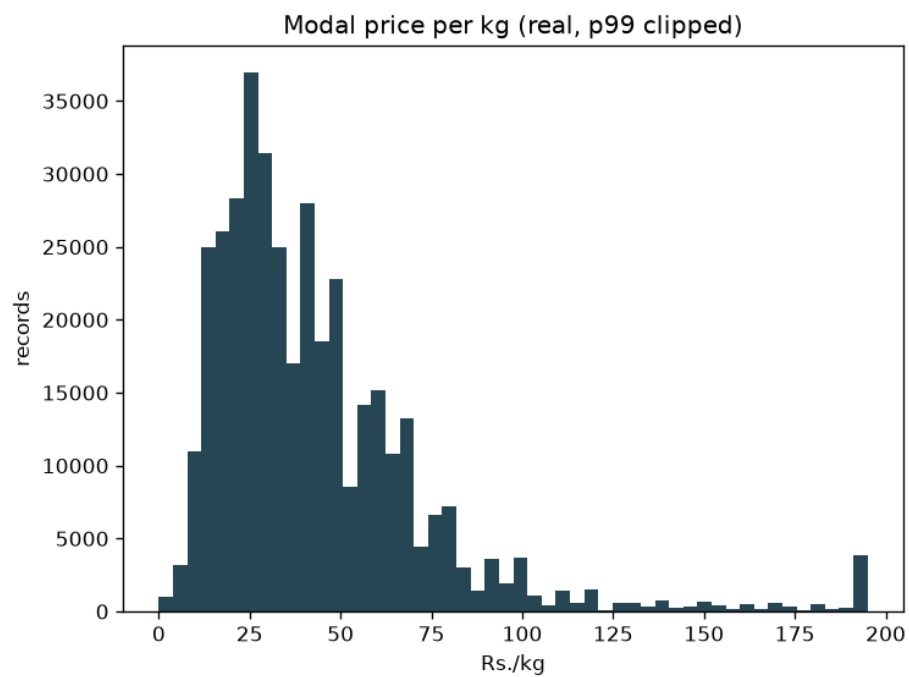


Fig 2 — Modal price/kg (p99 clipped). Observation: strong right skew — staples cluster low, a few high-value spices stretch the tail → per-kg standardisation + IQR caps justified.

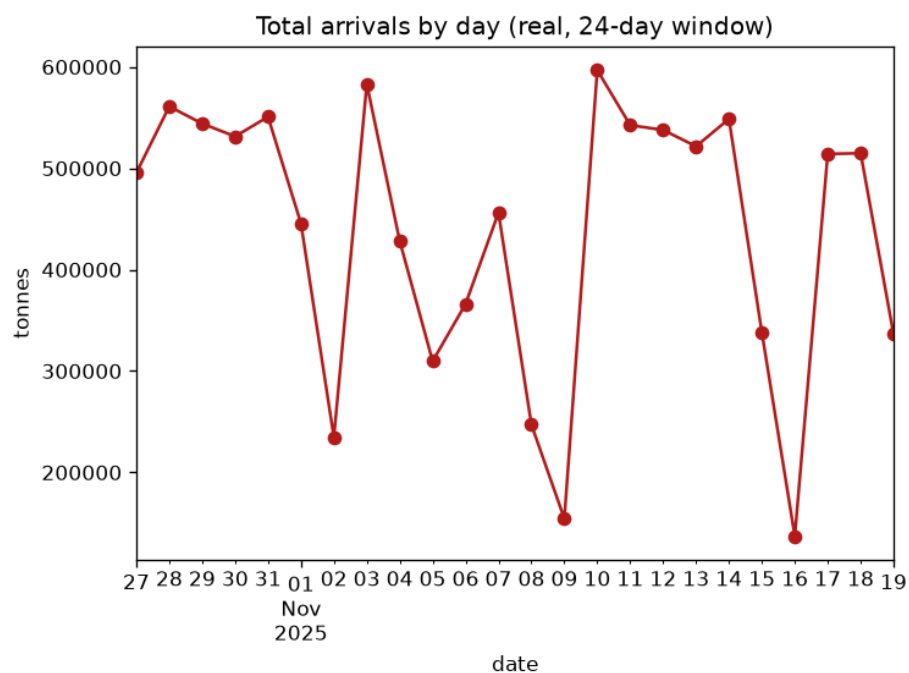


Fig 3 — Total arrivals per day over the 24-day window. Observation: wide day-to-day swings with gaps (not all mandis report daily) → no seasonality claim; rolling baselines use observed rows only.

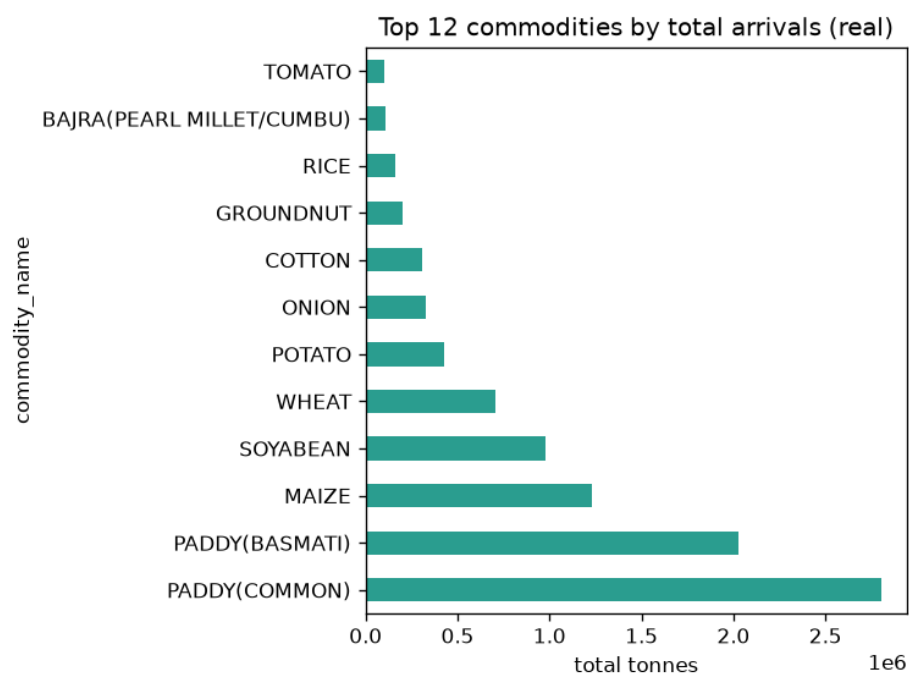


Fig 4 — Top 12 commodities by arrivals: Onion, Wheat, Potato, Tomato lead. Observation: perishables dominate → glut-spoilage framing fits; Tamil Nadu holds ~35% of rows → state dummies + per-kg features stop 'big state always wins'.

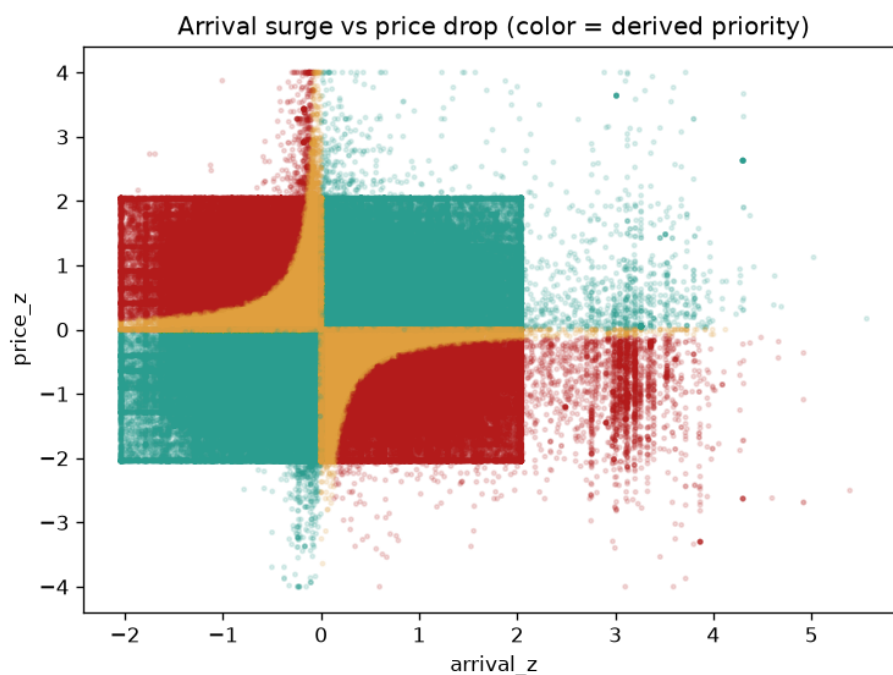


Fig 5 — Arrival surge (x) vs price drop (y), coloured by priority. Observation: High-priority points sit in the high-arrival + negative-price quadrant → validates the $S = \text{arrival_z} \times (-\text{price_z})$ rule.

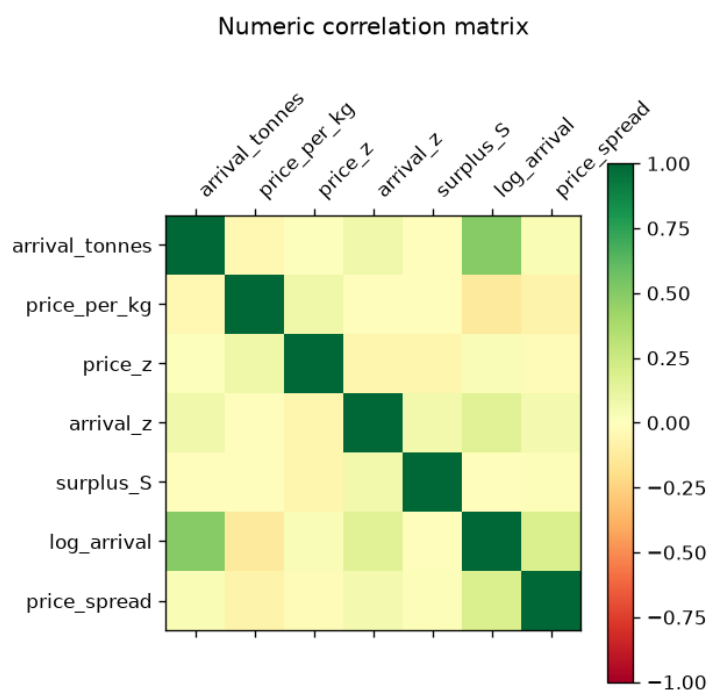


Fig 6 — Numeric correlations. Observation: S tracks $\text{arrival_z} / -\text{price_z}$ by construction; no hidden $|r| > 0.85$ among base numerics → low multicollinearity risk for CE-2.

8b. Supplementary EDA (bias checks & operations)

Three extra charts added after independent review. Frozen Figs 1–6 above are unchanged. Fig 7 answers the biggest bias worry; Fig 8 is reported honestly as a weak effect; Fig 9 is operational (where the first trucks go).

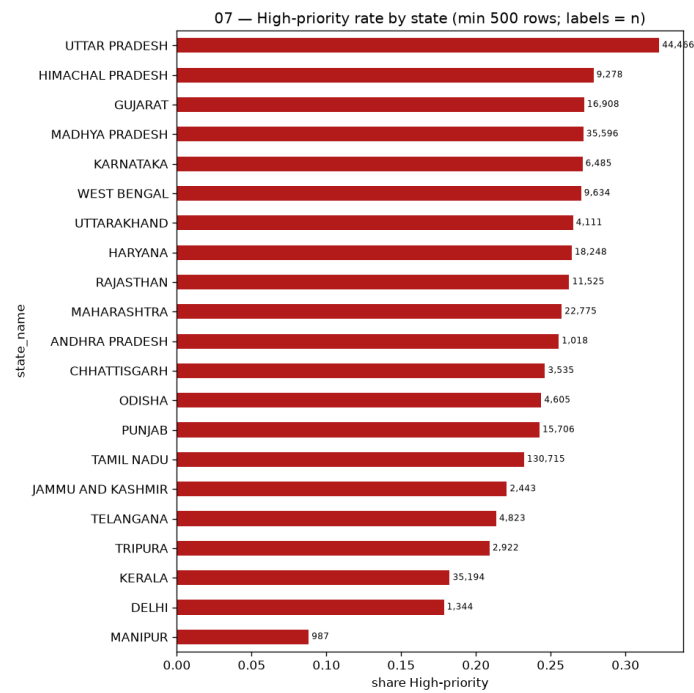


Fig 7 — High-priority rate by state (min 500 rows; labels = n). Observation: Tamil Nadu's High-rate is 0.232, below the 0.25 average, despite holding ~34% of rows → volume dominance does not drive labels; state dummies are controls, not a TN detector.

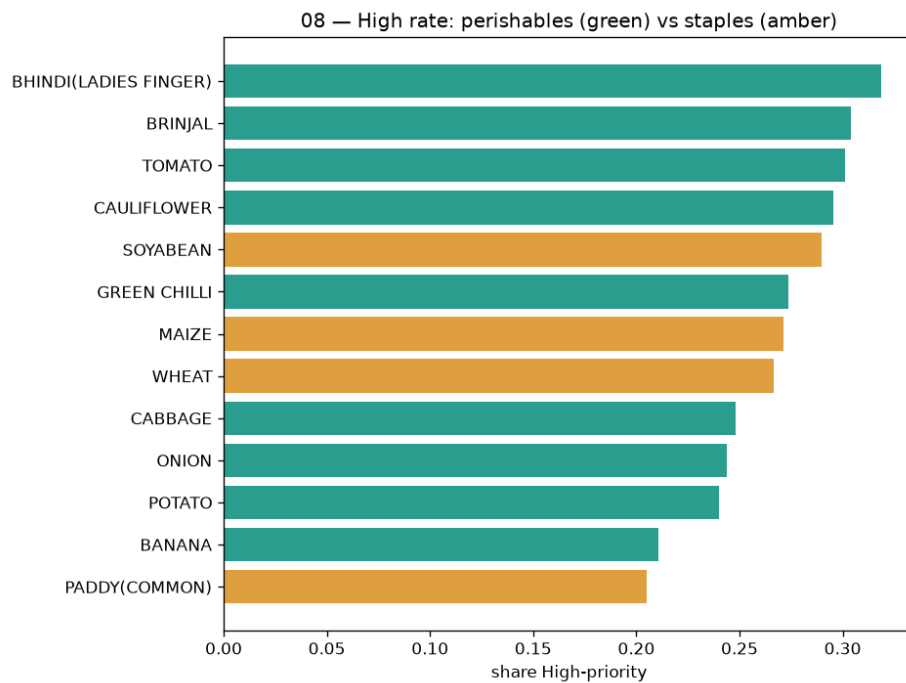


Fig 8 — High rate: perishables 0.270 vs staples 0.257. Observation: gap is small — consistent with the spoilage framing but not confirmation in a 24-day window; do not oversell.

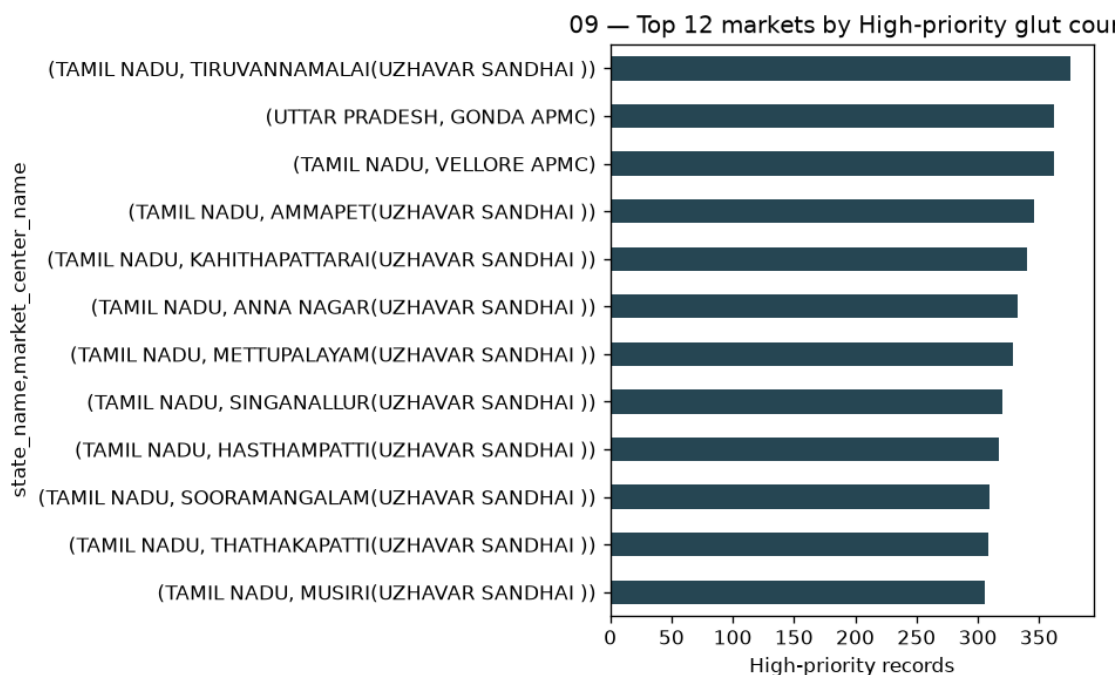


Fig 9 — Top 12 markets by High-priority glut count (Tiruvannamalai 376, Gonda 362, Vellore 362 lead). Observation: TN Uzhavar Sandhai outlets dominate by record count (many small direct-sale rows) → pair with tonnage before any dispatch claim.

9. Feature Selection / Engineering — X and y

Formula (from project empirical report): per market-commodity, 6-period rolling baselines (min 2 obs; commodity-median fallback), then $S = \text{arrival_z} \times (-\text{price_z})$. Label: High if $S \geq p75$ (0.3191), Medium if $S \geq p40$ (0.0), else Low. Thresholds frozen in `cleaning_report.json`.

#	Feature	Why
1–3	price_per_kg, arrival_tonnes, log_arrival	Core supply signal (log tames skew)
4–7	baseline_mean/std, arr_mean/std	Local norms per market-crop
8–10	price_z, arrival_z, surplus_S	Normalised shock + joint glut score (S drives y)
11–12	price_spread, geo_missing	Bargaining spread; missing-geo flag
13–16	month/day, is_weekend, lat/lon	Calendar + spatial (Haversine ranking in CE-2)
17–61	state + commodity dummies	Controls for region/crop effects

$y = \text{redistribution_priority}$ (derived triage proxy, disclosed — not observed waste). $X = 61$ columns above. Dropped from X: raw min/max/modal quintal (superseded by per-kg), market/district names (ranking keys, too many cards).

10. Pre-Processed Dataset Screenshot

Head of `data/processed/freshroute_foodbank_cleaned.csv` (383,753 × 22, missing = 0). Full table reproducible via the command in the spec sheet. Key columns shown:

date	state	market	commodity	arr t	Rs/kg	S	priority
2025-10-27	MADHYA PRADESH	A LOT APMC	GARLIC	26.59	5.00	3.04	High
2025-10-28	MADHYA PRADESH	A LOT APMC	GARLIC	10.71	15.00	0.50	High
2025-10-28	MADHYA PRADESH	A LOT APMC	GARLIC	10.71	5.00	-0.33	Low
2025-10-29	MADHYA PRADESH	A LOT APMC	GARLIC	32.25	5.00	0.55	High
2025-10-30	MADHYA PRADESH	A LOT APMC	GARLIC	31.53	5.00	0.38	High

date	state	market	commodity	arr t	Rs/kg	S	priority
2025-10-31	MADHYA PRADESH	A LOT APMC	GARLIC	19.69	10.00	0.14	Medium
2025-11-03	MADHYA PRADESH	A LOT APMC	GARLIC	20.60	5.00	-0.02	Low
2025-11-03	MADHYA PRADESH	A LOT APMC	GARLIC	20.60	30.01	0.48	High

Encoded: freshroute_foodbank_encoded.csv (383,753 × 62). Verify: script asserts 0 missing before saving.

11. Conclusion

CE-1 is complete on fully real, India-native data: messy 394k × 22 → clean 383,753 × 22 → encoded 383,753 × 62, frozen X/y, 6 EDA figures, disclosed S-rule, zero missing. Key finding: redistribution urgency separates on **arrival-surge × price-drop (S), not raw volume**. Logged biases: Tamil Nadu dominance, 24-day window (no seasonality), 37% geo-imputed, perishables over-represented. Dataset is ready for CE-2 model training.

12. Project Planning & Future Scope

CE-2: Logistic Regression / RandomForest / XGBoost on F1-macro with stratified split + per-state audit. Then the India pilot layer: (a) WFP long-baseline prices for seasonality, (b) Maharashtra PDS AAY/PHH demand layer (vulnerability V_d), (c) mandi → hub Haversine ranking with capacity, (d) volunteer dispatch + FSSAI-aligned logging. Field pilot required before any real-world impact claim. Reproduce everything with `pip install -r requirements.txt; python scripts/clean_dataset.py`.

Appendix CE-2. Next-Day Glut Forecast — Model Results

Task change (grilled decision): instead of classifying today's row (which would leak, since y is computed from S in X), the model **forecasts tomorrow**: features at day $t \rightarrow$ priority at day $t+1$ for the same market-commodity (352,498 pairs). Time split: train on target-dates before 2025-11-14 (267,804), test on the last 6 days (84,694).

Metric: F1-macro. Run: `python scripts/train_model.py`.

Model	F1-macro (test)	Accuracy (test)	Read
Majority baseline	0.181	0.373	Honesty anchor
Logistic Regression	0.304	0.384	Linear signal only; glut is nonlinear
RandomForest	0.516	0.549	Strong; bagging captures surge $\times$ drop
XGBoost (winner)	0.517	0.564	Best; $\sim 3\times$ baseline, picked for dispatch demo

Why modest scores are good news: a leaking model would print 0.95+. F1 ~ 0.52 on a 3-class next-day forecast from a 24-day window is an honest signal — and trees doubling the linear score justifies XGBoost. Named fix for higher scores: longer history (WFP baseline).

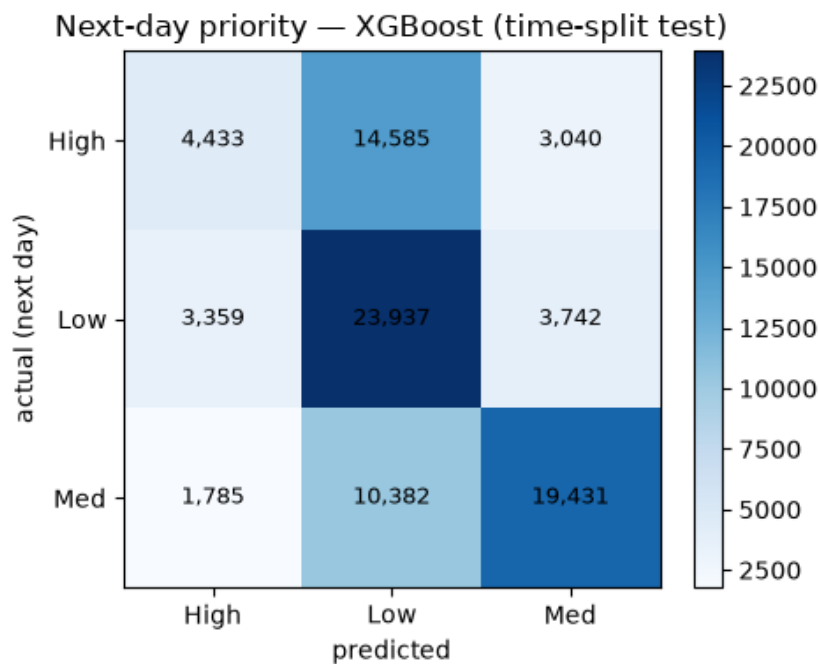


Fig 10 — Confusion matrix, XGBoost on unseen last-6-days (rows = actual next-day, cols = predicted). Per-class matrices for LogReg/RF saved alongside as `10_confusion_LogReg/RandomForest.png`.

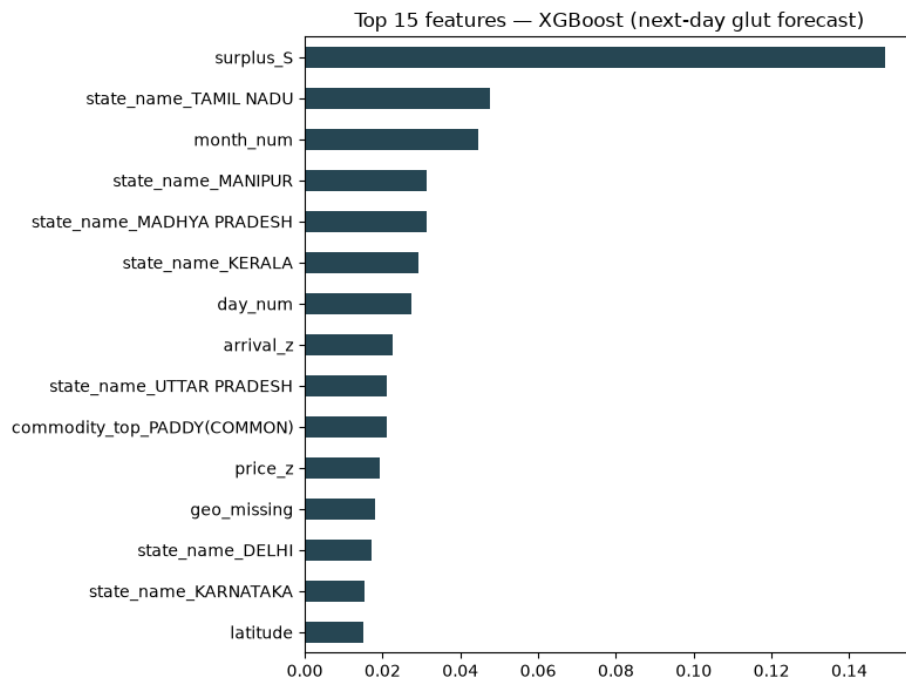


Fig 11 — Top 15 XGBoost features for next-day glut. Arrival/price dynamics and baselines dominate, as the S-formulation predicts.

Dispatch demo (optimizer layer)

Test-set rows predicted High are ranked by $P(\text{High}) \times \text{arrival tonnes}$ into a 50-row move-first list (data/processed/dispatch_list_demo.csv): top entries lead with Tiruvannamalai / Gonda / Vellore glut calls for 2025-11-14 onward. This is the classifier → ranking step the project title promises; capacities and hub distances plug in next.